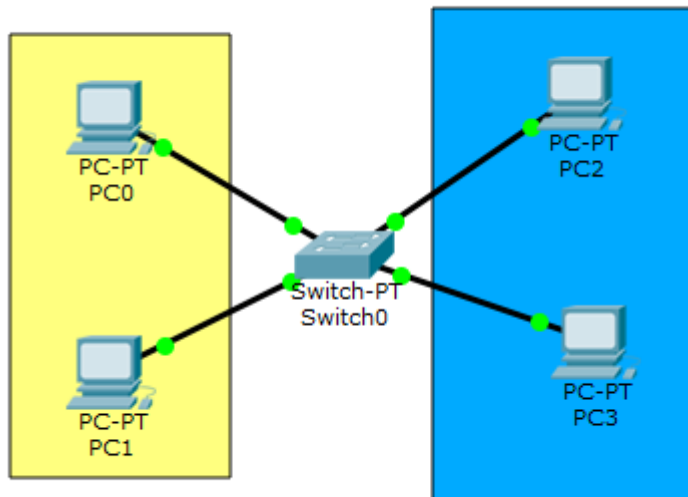




Lab1 – VLAN Configuration on Cisco Switches

We use Cisco Packet Tracer to configure VLANs on Cisco switches.

Let's consider the following diagram.



Objective: create two VLANs: *Marketing* and *Accounting*.

Step 1: Creating the topology and testing machine connectivity

- First, we create our topology using one switch and four computers.
- All four computers belong to the same network: 192.168.1.0/24.
- The computers have IP addresses ranging from 192.168.1.1/24 to 192.168.1.4/24.
- We test connectivity and ensure all computers can communicate with each other. Note the ping results.

Step 2: Creating VLANs and adding interfaces to VLANs

The goal is to have two computers in VLAN 2 and the other two computers in VLAN 3.

To create the VLANs, follow these commands:

1. First, change the switch name with the “hostname” command.

```
1 Switch>enable
2 Switch#configure terminal
3 Enter configuration commands, one per line. End with CNTL/Z.
4 Switch(config)#hostname SW2
5 SW2(config)#
```

2. Create the VLANs using the “VLAN” command.

1	SW2(config)#vlan 2
2	SW2(config-vlan)#name Marketing
3	SW2(config-vlan)#vlan 3
4	SW2(config-vlan)#name Accounting
5	SW2(config-vlan)#vlan 4

3. To verify that the VLANs have been created, type the command “do sh vlan” from configuration mode.

If everything went well, you should see a table showing the names of the VLANs created.

4. Now, assign interfaces to each VLAN.

Initially, all ports belong to VLAN 1 (the default VLAN). We assign the interfaces to the new VLANs using the “**switchport**” command.

1	SW2(config)#interface fastEthernet 0/1
2	SW2(config-if)#switchport mode access
3	SW2(config-if)#switchport access vlan 2
4	SW2(config-if)#exit
5	SW2(config)#interface fastEthernet 1/1
6	SW2(config-if)#switchport mode access
7	SW2(config-if)#switchport access vlan 2
8	SW2(config-if)#

We added [fastEthernet 0/1] and [fastEthernet 1/1] to VLAN 2, which is our *Marketing* VLAN. Do the same to add the other two interfaces to VLAN 3.

5. Finally, type “do sh vlan” again and **check the result**.

You will notice the interface numbers belonging to each VLAN.

6. **Test ping** between the different machines and note the results.

You should have intra-VLAN communication (within the same VLAN) but no inter-VLAN communication (between two different VLANs).